Additional Questions

1. Connection oriented Vs Connection less

Connected oriented service.	Connection less service.
1. Related to the telephone system.	1. Related to the Postal system.
2. Preferred by long & steady communication	2. Preferred by bursty communication.
3. Necessary	3. Not compulsory.
4. Feasible	4. Not feasible
5. Congenstion is not Possible.	5. Congenstion is Possible.
6. Gives guarantee of reliability.	6. Doesn't give the guarantee of reliability.
7. Packets follow the same route.	7. Packets doesn't follow the same route.

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2. Circuit Switching Vs Packet Switching.

Circuit switching	Packet switching
1. There are 3 Phases — (i) Connection establishment (ii) Data transfer (iii) Connection released	1. Data transfer takes place.
2. Each data unit know the entire Path address which is Provided by the source.	2. Each data unit know the final destination address, intermediate Path is decided by the routers.
3. Data is Processed at source system only.	3. Data is Processed at all the intermediate node including source System.
4. Delay between data units in Circuit Switching is uniform.	4. Delay between data units in Packet Switching is not uniform.
5. More reliable.	5. Less reliable.
6. Wastage of resources are more in Circuit switching.	6. Less wastage of resources
7. It is not a store & forward technique.	7. It is a store & forwared technique.
8. Transmission of data is done by the source.	8. transmission of data is done not only by the source, but also by the intermediate routers.
9. Congestion can occure dureing connection establishment time.	9. Congestion can occure dureing data transfer Phase.

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3. Virtual Circuit Vs Datagram.

Virtual Circuit	Datagram
1. Each Packet contains a short VC number.	1. Each Packet contains the Source & the destination address.
2. State information about each VC is maintained.	2. Doesn't hold Packet level state information.
3. Route is chosen when VC is setup.	3. Each Packet is routed independently.
4. Congestion control is easy if enough buffers can be allocated in advance.	4. Difficult.
5. Connection oriented service.	5. Connection-oriented & Connection less service

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4. Describe ATM. Property/Features, advantage of ATM.

ATM stands for Asynchronous Transfer Mode. An ATM is simply a data terminal with two input and four output devices. ATM technology uses ATM cell for data transmission between source & destination. ATM cell uses fixed size of 53 bytes which consists of header (5 bytes) & data (48 bytes). ATM transmits cells only when there is data to be transmitted unlike STM (Synchronous Transfer Mode) where bandwidth is assigned Periodically like TDM. ATM is connection oriented which uses Virtual Packet switching. Multiple logical connections are multiplexed over single Physical connection.

Properties :

1. The ATM standard defines a full suite of communication Protocols from the transport layer all the way down through the Physical layer.

2. It uses Packet switching with fixed length Packets of 53 bytes.

3. ATM uses Virtual Circuits (VCs)

4. Provides no retransmission on link-by-link basis.

5. Provides congestion control on an end-to-end basis.

6. ATM can run over Just about any Physical layer.

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Advantages:

1. It is optimized to transport voice, data & video.
2. Easy to integrate with LAN, MAN & WAN network types.
3. It is QoS oriented and high speed oriented.
4. It uses simplified network infrastructure.

5. What is Ethernet? Draw the frame format. What are the difference between fast and gigabit Ethernet?

Ethernet is the traditional technology for connecting wired LANs, enabling devices to communicate with each other via a protocol.

Frame format:

Preamble	SFD	Destination MAC	Source MAC	Type	Data & Pad	FCS
7 bytes	1 byte	6 bytes	6 bytes	2 bytes	46-1500 bytes	4 byte

Preamble: Informs the recieving system that a frame is starting & enables synchronization.

SFD (Start frame Delimiter): Signifies that the destination MAC address field begins with the next byte.

Destination MAC: identifying the ~~sending~~ recieving system.

Source MAC: identifies the sending system.

Type: define the type of protocol inside the frame. Ex- IPV4 or IPV6

Data and Pad: Contains the payload data.

FCS (Frame check sequence): Contains a 32-bit cyclic Redundancy Check (CRC) which allows detection of corrupted data.

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Basis for Comparision	Fast Ethernet	Gigabit Ethernet
Basic	offers 100 Mbps speed	Provide 1 Gbps speed
Delay	Generate more delay	Less comparitively
Configuration	simple	complicated
Coverage	Can cover distance up to 40km.	Has the limit of 70km
Relation	Successor of 10-Based-T ethernet	A successor of fast ethernet
Round trip delay	100-500 bit times	4000 bit times

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6. Why CSMA/CD & CSMA/CA are used? How do they avoid colision?

CSMA/CD: is used to improve CSMA Performance by terminating transmission as soon as a colision is detected.

CSMA/CA: is particularly important for wireless networks, where the colision detection of the alternative CSMA/CD is not possible due to wireless transmitters desensing their recieverer during Packet transmission.

Avoidance of colision:

CSMA/CA increases network traffic as it requires sending out a signal to the network even before transmitting any real data.

CSMA/CD works by detecting a colision in the medium and backing off as necessary. Assume, we have a LAN with a number of hosts. When one host wants to transmit a frame, it listens on the medium to check if the medium is busy.

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7. Distinguish between 1 Persistent CSMA, P- Persistent & non-Persistent CSMA.

1 Persistent	p Persistent	non Persistent
1. If the channel is busy, the station waits until it becomes idle.	1. If the channel is busy, it wait for fixed interval of time.	1. If channel is busy, station waits until next time slot.
2. When the station detects an idle-channel, it immediately transmit the frame with probability 1.	2. If the channel is idle, it sends immediately	2. If channel is idle, it transmits with a probability P.
3. Station that wants to transmit data continiously senses the channel to check whether the channel is idle or busy.	3. The station doesn't continuously sense the channel.	3. With the probability $q = 1 - P$, the station then waits for the beginning of the next time slot.
4. No concept of time slot.	4. No concept of time slot	4. Data transmission occurs depending on fixed time slot.
5. Highest chance of colision	5. Reduces the chance of colision.	5. Reduces the chance of colision.
6. The propagation delay time affects this protocol. Thus inefficient.	6. It reduces the efficiency of the network	6. Improves the efficiency of the network.

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